LAB 04

TITLE

Docker, Python and Database Connections (MySQL and MongoDB)

Name:

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Registration number:

(2023-BSCS-44)

Section:

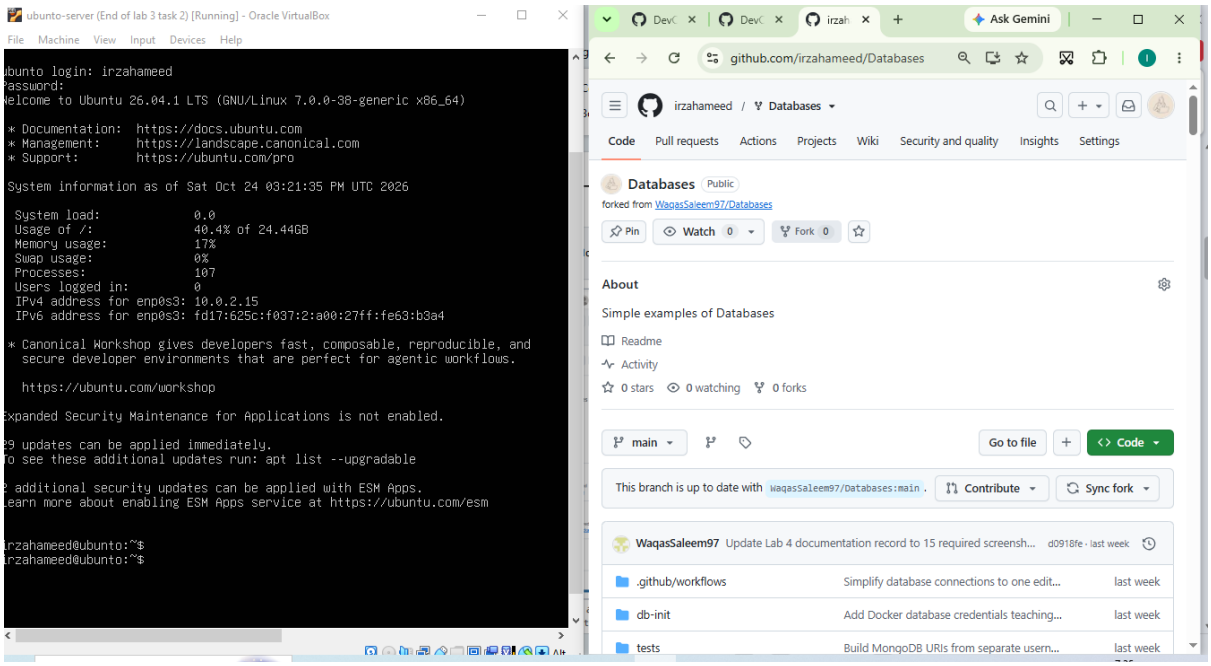
A

GitHub username:

irzahameed

Task 1: Fork and Clone the Databases Repository

databases_fork.png



databases_clone.png

```
irzahameed@ubunto:~$ git clone git@github.com:irzahameed/Databases.git
Cloning into 'Databases'...
remote: Enumerating objects: 101, done.
remote: Counting objects: 100% (101/101), done.
remote: Compressing objects: 100% (90/90), done.
remote: Total 101 (delta 30), reused 1 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (101/101), 39.86 KiB | 251.00 KiB/s, done.
Resolving deltas: 100% (30/30), done.
irzahameed@ubunto:~$ cd Databases
irzahameed@ubunto:~/Databases$
```

Task 2: Install Docker, Docker Compose, and Python on Ubuntu

docker_installation.png

```
Last login: Sun Oct 25 15:02:55 2026 from 10.0.2.2
irzahameed@ubunto:~$ git --version
git version 2.53.0
irzahameed@ubunto:~$ docker --version
Docker version 29.1.3, build 29.1.3-0ubuntu4.1
irzahameed@ubunto:~$ docker compose version
Docker Compose version v5.6.1
irzahameed@ubunto:~$
```

Task 3: Start the Database Services

database_services.png

```
(.venv) irzahameed@ubunto:~/Databases$ docker compose ps
NAME                    IMAGE          COMMAND                  SERVICE      CREATED         STATUS                   PORTS
databases-mongo-dev-1   mongo:8.0.4    "docker-entrypoint.s…"   mongo-dev    4 minutes ago   Up 4 minutes (healthy)   127.0.0.1:27017->27017/tcp
databases-mongo-prod-1  mongo:8.0.4    "docker-entrypoint.s…"   mongo-prod   4 minutes ago   Up 4 minutes (healthy)   127.0.0.1:27019->27017/tcp
databases-mongo-test-1  mongo:8.0.4    "docker-entrypoint.s…"   mongo-test   4 minutes ago   Up 4 minutes (healthy)   127.0.0.1:27018->27017/tcp
databases-mysql-1       mysql:8.4      "docker-entrypoint.s…"   mysql        4 minutes ago   Up 4 minutes (healthy)   33060/tcp, 127.0.0.1:3307->3306/tcp
(.venv) irzahameed@ubunto:~/Databases$
```

database_ports.png

```
(.venv) irzahameed@ubunto:~/Databases$ docker compose port mysql 3306
127.0.0.1:3307
(.venv) irzahameed@ubunto:~/Databases$ docker compose port mongo-dev 27017
127.0.0.1:27017
(.venv) irzahameed@ubunto:~/Databases$ docker compose port mongo-test 27017
127.0.0.1:27018
(.venv) irzahameed@ubunto:~/Databases$ docker compose port mongo-prod 27017
127.0.0.1:27019
(.venv) irzahameed@ubunto:~/Databases$
```

Task 4: MySQL with Values in the Code

mysql_hardcoded_connection.png

```
(.venv) irzahameed@ubunto:~/Databases$ nano db_connect.py
(.venv) irzahameed@ubunto:~/Databases$ python db_connect.py
Connected to MySQL: school_mysql
Server: 127.0.0.1 3307
Environment: dev
```

Task 5: MongoDB with Values in the Code

mongo_hardcoded_connection.png

```
(.venv) irzahameed@ubunto:~/Databases$ nano db_connect.py
(.venv) irzahameed@ubunto:~/Databases$ python db_connect.py
Connected to MongoDB: school_dev
Server: ('127.0.0.1', 27017)
Environment: dev
```

Task 6: MongoDB with os.getenv()

mongo_env_dev.png

```
(.venv) irzahameed@ubunto:~/Databases$ python db_connect.py
Connected to MongoDB: school_dev
Server: ('127.0.0.1', 27017)
Environment: dev
(.venv) irzahameed@ubunto:~/Databases$
```

mongo_env_test.png

```
(.venv) irzahameed@ubunto:~/Databases$ python db_connect.py
Connected to MongoDB: school_test
Server: ('127.0.0.1', 27018)
Environment: test
(.venv) irzahameed@ubunto:~/Databases$
```

mongo_env_prod.png

```
(.venv) irzahameed@ubunto:~/Databases$ python db_connect.py
Connected to MongoDB: school_prod
Server: ('127.0.0.1', 27019)
Environment: prod
(.venv) irzahameed@ubunto:~/Databases$
```

Task 7: MongoDB with a Credentials File

credential_files_created.png

```
(.venv) irzahameed@ubunto:~/Databases$ cp -n credentials.ini.example credentials.ini
cp: warning: behavior of -n is non-portable and may change in future; use --update=none instead
(.venv) irzahameed@ubunto:~/Databases$ ls credentials.ini
credentials.ini
(.venv) irzahameed@ubunto:~/Databases$
```

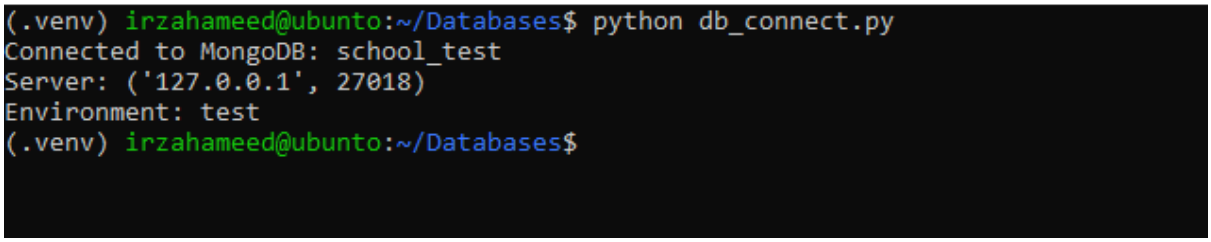
credential_file_permissions.png

```
(.venv) irzahameed@ubunto:~/Databases$ chmod 600 credentials.ini
(.venv) irzahameed@ubunto:~/Databases$ stat -c '%a %n' credentials.ini
600 credentials.ini
(.venv) irzahameed@ubunto:~/Databases$
```

mongo_file_dev.png

```
(.venv) irzahameed@ubunto:~/Databases$ python db_connect.py
Connected to MongoDB: school_dev
Server: ('127.0.0.1', 27017)
Environment: dev
(.venv) irzahameed@ubunto:~/Databases$
```

mongo_file_test.png



mongo_file_prod.png

